

StormHacks Compliance Navigator: PM Handoff

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1. Overview

We are building a compliance navigator that tells a new BC sole proprietor exactly which registrations, tax accounts and employer obligations apply to them, why, and what to do next, with every claim linked to an official government source.

Product in one sentence: a business owner answers a few questions, three specialist AI agents check their situation against official BC and federal sources, and they get a readiness score, cited explanations, upcoming triggers, and direct links to act.

Who it is for (MVP persona)

- **Structure:** unincorporated sole proprietor, the default for most people starting out.
- **Stage:** pre-launch through the first year, when nearly every registration decision happens for the first time.
- **Size:** solo, or about to make a first hire or two.
- **Business type:** product or service businesses run from home, online, or both (handmade goods shop, freelance designer, tutor, photographer, small online store).
- **Situation:** no lawyer, accountant or HR person. They piece things together from government websites, Reddit and friends.

Jurisdiction: City of Vancouver, Province of BC, and federal (CRA) rules that apply to a Vancouver-based sole proprietor.

The three legal areas in scope

1. Registration and licensing: "Am I allowed to exist and operate here?"
2. Tax registration: "Do I have to charge tax, and when?"
3. Employer obligations: "What changes the day I hire someone?"

Why we win against what exists today

Alternative	What it does	What it misses
BizPaL	Lists permits and licences that may apply, with links	Does not say what does apply, no next steps, no triggers
Ownr	Registers a sole proprietorship name (\$49 + GST)	Registration only; GST registration in Ownr is for corporations only

Small Business BC checklist	Generic startup checklist	Not personalised, no triggers
DIY (gov sites, Reddit, friends)	Free	Scattered, mixed quality, no one flags thresholds
Accountant	Tax and payroll advice	Costly; rarely covers licensing; only knows what you tell them

We are the only option that covers all three areas, says what applies and why, cites the official source, gives the next action with a direct link, and flags new triggers (first hire, \$10,000 PST line, \$30,000 GST line).

Hackathon prize tracks we are targeting

Track	How we qualify	Owner
TiDB x AI Open Build	TiDB vector search (and full-text/hybrid if region allows) powers retrieval; agent traces stored in TiDB	Developer 1
MLH Best Use of Gemini API	Gemini Flash runs intake, the three agents, chat and drafts; Gemini embeddings	Developer 2
SSSS Python track	Entire backend is Python (FastAPI)	Both devs
MLH Best .Tech Domain	Register a .tech domain for the deployed app	PM
Enactus UNSDG track	Pitch framing: SDG 8.3 (small business formalization), 16.3 (access to justice), 17 (built on government open data)	Business
IATSU Best Design	Figma design turned into the front end	Designer

Check StormHacks rules on how many sponsor tracks one project can enter and whether pre-event designs or research are allowed.

2. Business logic (the most important section)

The product is only trustworthy if these rules are implemented exactly as written. Code decides what applies and calculates the score; AI only explains, cites and flags. The research owner must validate every rule below against official sources before it is coded.

2.1 Registration and licensing: "Am I allowed to exist and operate here?"

- **Business name registration (BC Registries):** required only if the owner trades under

a name other than their own legal name. Provincial registration fee is \$40, after a name approval request.

- **City of Vancouver business licence:** required to operate in Vancouver. Research owner must confirm how this applies to home-based and online-only businesses.
- **CRA business number:** needed before opening a GST account or a payroll account.
- **Why it matters:** everything else (bank account, tax registrations, operating legally) sits on top of this. It is also the cheapest stage to fix, roughly \$150 in total fees (research owner to confirm the Vancouver licence fee before we quote this).

2.2 Tax registration: "Do I have to charge tax, and when?"

This is where new businesses most often go wrong, because BC has two sales taxes with different triggers.

Tax	Trigger	Key details
BC PST (provincial)	Applies almost immediately to anyone selling taxable goods	Small seller exemption only if BOTH: gross sales of \$10,000 or less over the past 12 months (rolling window, not calendar year) AND no established business premises. Some seller types never qualify.
GST (federal)	Must register once over \$30,000	Test is \$30,000 in a single calendar quarter or over four consecutive quarters (research owner to confirm wording against CRA). Voluntary registration earlier is allowed.

Rules the engine must respect:

- **Goods vs services matters.** PST mostly applies to goods; most services in our persona (designer, tutor, photographer) are not PST-taxable. As of October 1, 2026, PST expanded to certain professional services (accounting, architecture, engineering and others). The tax logic must ask "goods, services or both?" first.
- **Rolling windows.** One busy season can end PST small seller status. Calculators must use monthly revenue, not a single annual number.
- **Gray area to flag, not resolve:** regular sales at markets or fairs may affect the "no established premises" condition. Flag for review.

2.3 Employer obligations: "What changes the day I hire someone?"

Hiring the first person is the biggest single jump in legal responsibility. On day one, these

switch on:

1. WorkSafeBC registration.
2. A CRA payroll account (requires a business number).
3. Minimum wage of \$18.25/hour (in effect since June 1, 2026).
4. A pay statement every payday.
5. Payroll records kept for the required period (we have stated four years; research owner must confirm the exact rule).

Gray area to flag: whether someone is an employee or a contractor. The agent explains the question and recommends checking; it never decides.

2.4 Rules that apply across everything

- **Unknown is never false.** If a fact is missing (for example revenue), the requirement is "undetermined" and we ask for it. We never assume the owner is fine.
- **Every requirement has a timing:** `now` (required today), `trigger` (switches on at a threshold or event), or `recommendation` (good practice, never counted in the score).
- **Dependencies set the order of next steps:** name registration, then business number, then GST or payroll accounts.
- **Links come only from the registry.** The AI never invents a URL, fee, deadline or form.
- **No official evidence, no claim.** If retrieval finds nothing, we say "we couldn't find an official source for this" and suggest who to ask.
- **We give legal information, not legal advice.** A disclaimer is shown, and gray areas route to a professional.

2.5 How the readiness score works

The score answers: "Of the things legally required of you today, how many have you done?"

- Only `required_now` legal obligations count toward the score.
- `upcoming` triggers, recommendations and undetermined items are shown separately and never lower the score.
- Each requirement has a priority weight: high = 3, medium = 2, low = 1.
- Status credit: done = 1.0, in progress = 0.5, not done = 0.
- Score = $100 \times (\text{sum of weight} \times \text{credit}) / (\text{sum of weights})$, calculated per area and overall.
- Same answers always produce the same score. This is our main answer to "isn't this just ChatGPT?"

3. User workflow and demo persona

The owner goes from a short intake to a scored, cited action plan in one flow, and any confirmed change to their facts re-runs the whole assessment.

3.1 Step by step (what the owner experiences)

- 1. **Intake.** The owner fills a short form and optionally writes one or two sentences about their business.
- 2. **Confirm facts.** The system turns the description into proposed facts (for example "sells: both goods and services"). The owner confirms or edits each one.
- 3. **Follow-up questions.** If facts are missing (monthly revenue, trading name, hiring plans), the system asks targeted questions. The owner can skip; skipped facts stay "unknown".
- 4. **Assessment.** The three agents run and the owner sees a readiness score.
- 5. **Dashboard: now, next, later.**
 - Now: what is required today, each with a "Go here" link. This is what the score measures.
 - Next: approaching triggers ("you are at \$8,200 of the \$10,000 PST line; estimated to cross around March").
 - Later: what switches on when you hire.
- 6. **Why?** Each card expands to show the official source passage and link, plus the agent trace.
- 7. **Chat.** The owner asks questions; answers are grounded in official sources and cited.
- 8. **Actions.** Checklists, official links and inquiry email drafts. The owner sends drafts themselves; nothing is sent automatically.
- 9. **Update.** If the owner mentions a change in chat ("I'm hiring help next month"), the system proposes a fact update. Once confirmed, a new profile version is saved and the assessment re-runs.

3.2 Demo persona: Maya, "Wick & Co"

Maya makes candles in her Vancouver apartment and sells on Etsy and at weekend markets. One persona is used end to end so the story stays simple.

Demo moment	What happens on screen	Area
Trading name "Wick & Co" differs from her legal name	Name registration appears under Now, with the BC Registries link	Registration
Home-based in Vancouver	Business licence appears under Now (or flagged, depending on research findings)	Registration

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	Flagged, depending on record management,	
About \$8,200 in sales over the past 12 months, growing	PST small seller status holds today; PST registration shows under Next with an estimated crossing month	Tax
Regular weekend market sales	Flagged for review: may affect the "no established premises" condition	Tax
Well under \$30,000	GST shown as not yet required; voluntary registration shown as a recommendation	Tax
Mid-demo in chat: "I'm hiring help for the holidays"	Fact update proposed, confirmed, re-run: the five employer obligations move from Later to Now and the score updates	Employer

The hiring moment is the climax of the demo: "five obligations switch on in one day", happening live.

4. System architecture and tech stack

One Python backend, one TiDB database and one Gemini model family; the only AI calls are intake, the three agents, chat and email drafts.

4.1 Tech stack

Layer	Choice	Notes
Language	Python 3.11	Qualifies for the Python track
API	FastAPI + Uvicorn, Pydantic v2	All routes and data contracts
Database	TiDB Cloud Starter	Relational data, vectors, full-text; one store only
DB access	SQLAlchemy 2 + PyMySQL, or the pytidb SDK for search	Pin pytidb version; its API changes quickly
LLM	Gemini Flash via google-genai	Model name in config; check current name on the day
Embeddings	gemini-embedding-001	One fixed model and dimension; 2,048-token input limit per chunk
Ingestion	httpx , BeautifulSoup, lxml , pypdf	Approved sources only
Tests	Pytest + FastAPI test client	Acceptance tests in section 10
Front end	Figma design, built by the designer	Calls the FastAPI routes in section 8

	(React, TypeScript, Vite, Tailwind)	
Hosting	Backend on Render, Railway or Fly.io; front end on Vercel	<code>yourname.tech</code> to front end, <code>api.yourname.tech</code> to backend via CNAME

Do not add LangChain, LangGraph, Celery, Redis, a second vector database or separate microservices.

TiDB region warning: full-text search (needed for hybrid search) is only available in certain TiDB Cloud Starter regions. Create the cluster in a supported region on day one and confirm the feature works. Vector search alone still qualifies for the TiDB track.

4.2 TiDB data model

Table	What it holds	Owner
<code>sources</code>	One row per official page or document: URL, authority, title, retrieved date, version, content hash	Developer 1
<code>chunks</code>	Section-sized text pieces with metadata, embedding vector, full-text index	Developer 1
<code>requirements</code>	The registry: every obligation with its rules, timing, dependencies, links (section 7)	Developer 1 (content approved by research owner)
<code>business_profiles</code>	Versioned facts per business; each fact has value + confirmed flag	Developer 2
<code>revenue_entries</code>	Monthly revenue amounts per business, used by tax calculators	Developer 2 writes, Developer 1 reads
<code>assessments</code>	One row per run, tied to a profile version	Developer 1
<code>findings</code>	One row per requirement per assessment: status, explanation, cited chunk IDs, flags, confidence	Developer 2 writes, Developer 1 scores
<code>agent_runs</code>	Every agent tool call and result (the "agent trace")	Developer 2
<code>conversations</code>	Chat history: question, rewritten query, retrieved chunk IDs, claims, sources cited, proposed fact updates	Developer 2

4.3 Repository structure

backend/app/contracts/	Shared Pydantic models (retrieval, findings, profile)
backend/app/knowledge/	Dev 1: ingestion, chunking, embeddings, retrieval
backend/app/assessment/	Dev 1: applicability rules, calculators, scoring, sequencing
backend/app/intake/	Dev 2: fact extraction from free text
backend/app/agents/	Dev 2: orchestrator, base agent, three agents, tools
backend/app/chat/	Dev 2: grounded answers, citation validation
backend/app/actions/	Dev 2: checklists, inquiry drafts
backend/app/api/	Dev 2: FastAPI routes
backend/app/core/	Shared settings, DB setup
backend/data/registry/	Requirement registry (CSV/JSON, reviewed)
backend/scripts/	Ingestion and registry loader scripts
backend/tests/	Tests per module
contracts/examples/	Sample JSON requests and responses for the front end
frontend/	Designer-owned interface

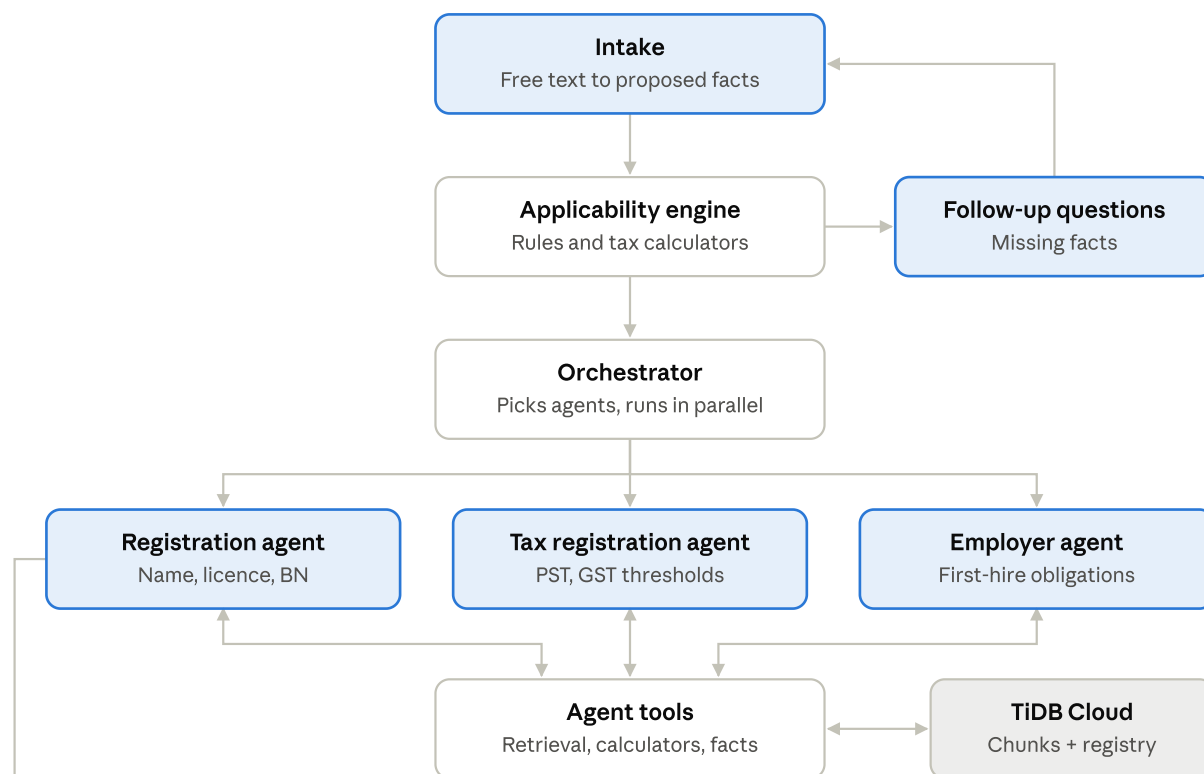
5. AI design: intake, engine, orchestrator and three agents

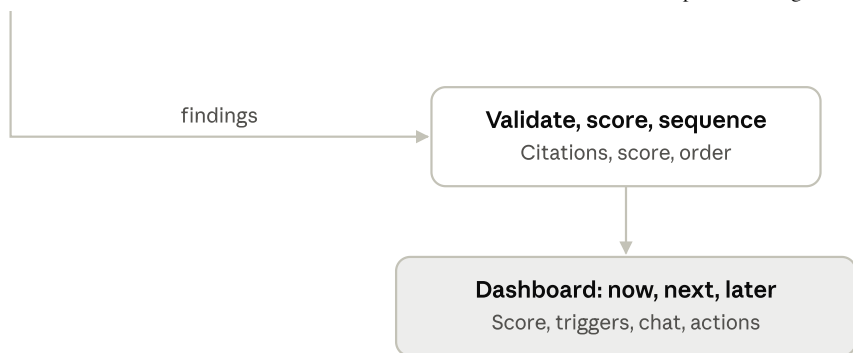
Code decides what applies and how the score is calculated; three specialist agents read official evidence, explain it to the owner, and flag gray areas.

Code decides what applies; three agents explain and cite

☒ Gemini, Developer 2

☐ Code, no AI, Developer 1

☐ Data and interface




A confirmed fact change re-runs the flow from intake

Tinted boxes call Gemini; outlined boxes are deterministic code. Agents only reach TiDB through the shared tools, and their findings go straight to validation and scoring.

5.1 Who decides what

Decision	Decided by	Why
Does a rule apply? Is a threshold crossed?	Code (Developer 1)	Must be exact and repeatable
Turning a free-text description into facts	Gemini intake call; owner confirms	Messy language is what LLMs are good at
Which evidence supports a requirement, and what it means for this owner	Agent (Gemini + tools)	Reading and explaining legal text
Gray areas (employee vs contractor, home-based licence edge cases, market stalls)	Agent flags; never resolves	Honest handling of legal uncertainty
The score and the order of steps	Code (Developer 1)	Same answers always give the same score
Links, fees, next actions	Registry only	No invented URLs or numbers

5.2 Intake (Developer 2)

- One Gemini Flash call with structured output (Pydantic schema) turns the owner's description into proposed facts, each with a confidence score.
- Example: "I sell candles on Etsy and do custom design work" proposes `sells = both, online = true`.
- The owner confirms or edits every proposed fact. Unconfirmed facts are never used as true.

5.3 Applicability engine and calculators (Developer 1, no AI)

The engine compares each requirement's `applies_if` rule with the confirmed facts and labels it `required_now`, `upcoming`, `not_applicable` or `undetermined` (with the list of missing facts).

Calculators are plain Python functions inside the engine:

- `pst_small_seller_test(profile)` : sums revenue over the rolling 12 months and checks the premises condition. Returns `exempt`, `must_register` or `undetermined`, plus the numbers used.
- `gst_small_supplier_test(profile)` : checks single-quarter and four-consecutive-quarter totals against \$30,000.
- `project_crossing(profile, threshold)` : straight-line projection from recent months to estimate when a threshold will be crossed. Always labelled as an estimate.

Missing facts become follow-up questions (Developer 2 writes the wording). This loops until the owner has answered what they can.

5.4 Orchestrator (Developer 2, plain Python, not an LLM)

- **Registration agent:** always runs.
- **Tax registration agent:** always runs ("you are under the threshold" is useful too).
- **Employer agent:** full mode if `has_employees = true`; pre-hire mode ("what changes the day you hire") if `plans_to_hire` is true or unknown; skipped if the owner said no.
- Runs active agents in parallel with `asyncio`, collects findings, removes duplicates, and passes them to validation and scoring.

5.5 How each agent works

Each agent is a Python class with:

- **Scope:** only requirements whose `area` matches (registration, tax or employer).
- **System prompt:** written as that specialist, listing its known gray areas.
- **Tools:** Python functions Gemini can call (table below).
- **Step limit:** about 6 tool calls per run, so it cannot loop forever.

Tool	Registration	Tax	Employer	Provided by
<code>retrieve_evidence(requirement_id, query)</code>	Yes	Yes	Yes	Developer 1 retrieval service
<code>get_requirement(id)</code>	Yes	Yes	Yes	Developer 1 registry

<code>get_profile_fact(key)</code>	Yes	Yes	Yes	Developer 2 profile service
<code>get_calculator_result(name)</code>	No	Yes	No	Developer 1 calculators
<code>flag_for_review(requirement_id, reason)</code>	Yes	Yes	Yes	Developer 2

Two-phase run (per agent):

1. **Investigate:** Gemini Flash gets the agent's scope and applicability results, then calls tools: pulls evidence for each requirement, reads calculator outputs, checks facts, flags anything ambiguous. The `google-genai` SDK can run Python functions as tools automatically; cap the number of calls.
2. **Report:** a second Gemini call with structured output turns what it gathered into one finding per requirement.

Splitting the phases keeps tool use and strict JSON output from conflicting, and makes each phase easy to debug.

Finding schema (agent output):

```
class Finding(BaseModel):
    requirement_id: str
    status: Literal["done", "in_progress", "not_done", "not_yet_required",
"undetermined"]
    explanation: str                # plain English, for the owner
    claims: list[Claim]            # each claim cites source chunk IDs
    flags: list[str]               # gray areas sent for review
    confidence: float              # below ~0.6 becomes "check with a professional"
```

Agent trace: every tool call and result is saved to `agent_runs`. The UI shows it as a panel ("Tax agent read PST bulletin, ran the small seller test, flagged market stall sales"). This makes the agentic work visible to judges.

5.6 Agent specialisations

Agent	Owns	Typical gray areas it flags
Registration	Name registration, Vancouver business licence, CRA business number	Home-based or online-only licensing; using own name with a tagline
Tax registration	PST registration, GST registration, voluntary GST, charging tax correctly once registered	Goods vs services split; market stalls and the premises condition; revenue data missing

Employer	WorkSafeBC, payroll account, minimum wage, pay statements, payroll records, employee vs contractor	Contractor vs employee; family members helping out
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5.7 Validation, scoring and sequencing

- **Citation validation (Developer 2):** drop any claim citing a chunk the agent did not retrieve in this run, and any finding outside the agent's scope.
- **Scoring (Developer 1):** formula in section 2.5; only `required_now` obligations count.
- **Sequencing (Developer 1):** sort next actions by `depends_on` first, then by priority.

5.8 Chat and actions (Developer 2)

- A question is routed to the matching agent (a small classifier call or keyword rules), which uses the same tools and validation as the assessment, so chat and dashboard never disagree.
- If retrieval returns `insufficient_evidence`, the answer says so and suggests who to ask.
- If the owner mentions a new fact, the agent proposes a profile update; the owner confirms; the assessment re-runs.
- Inquiry email drafts use the company profile and the specific requirement. Shown with Copy and an "Open in email" link; never sent automatically.

5.9 Gemini usage and limits

- Gemini Flash is sufficient for intake, agents, chat and drafts; accuracy comes from the registry, retrieval and code-based scoring.
- Main risk is rate limits, not quality: three agents in parallel, each making several calls. Check quota, consider paid credits, and pre-run and cache the demo persona's assessment as a fallback.
- Keep temperature low for agents and validation-sensitive calls.

6. Knowledge base and RAG

About 10 approved official sources are cleaned, split by section, embedded and stored in TiDB; agents and chat only ever answer from what this retrieval returns.

6.1 How RAG works here

1. Official sources are collected and cleaned.
2. Each is split into section-sized chunks with metadata.

3. Each chunk is embedded with one fixed Gemini embedding model.
4. Chunks and vectors are stored in TiDB.
5. At run time, an agent asks the retrieval service for evidence for a specific requirement.
6. Retrieval filters by jurisdiction, requirement and date, then searches (vector, or hybrid if available).
7. It returns evidence chunks, or `insufficient_evidence`. It never returns a legal conclusion.

6.2 Approved sources (research owner finalises)

Area	Source	Authority
Registration	Proprietorship registration package and name request pages	BC Registries
Registration	Business licence pages, including home-based business rules	City of Vancouver
Registration	Business number page	CRA
Tax	PST registration pages and Small Sellers bulletin (PST 003)	BC Ministry of Finance
Tax	GST registration and small supplier pages	CRA
Employer	Employer registration pages	WorkSafeBC
Employer	Employment Standards Act sections on pay statements and records	BC Laws
Employer	Minimum wage page	Province of BC
Employer	Payroll account page	CRA
All	OneStop Business Registration overview (registers business, GST, PST, WorkSafeBC and payroll in one flow)	Province of BC

Source rules:

- Official government, regulator or legislation sources only. Blogs, Reddit, law-firm summaries and AI summaries can help find topics but are never the authority.
- Every source keeps: URL, authority, title, retrieved date, version or effective date, section heading, original text.
- Do not bulk-scrape CanLII; its terms prohibit it. BC Laws offers an official XML API

and an open licence.

6.3 Ingestion and chunking (Developer 1)

- **BC Laws:** fetch statute XML through the CiviX document API (add `/xml` to the document URL), parse sections with `lxml`.
- **Government web pages:** fetch the approved URL list with `httpx`, strip navigation with BeautifulSoup, split by heading.
- **Chunk by legal structure** (Act, part, section, or page heading), not fixed token sizes. Keep each chunk well under the 2,048-token embedding limit; split long sections at subsections and repeat the section header.
- **Target:** about 40 to 80 meaningful chunks.

6.4 Chunk metadata

`source_id`, `chunk_id`, `authority`, `jurisdiction_ids` (`CA`, `CA-BC`, `CA-BC-VANCOUVER`), `area`, `document_type`, `section_path`, `source_version`, `effective_from`, `effective_to`, `retrieved_at`, `review_status`, `requirement_ids`, `embedding_model`.

6.5 Retrieval contract (Developer 1 exposes, Developer 2 consumes)

```
class RetrievalRequest(BaseModel):
    query: str
    jurisdiction_ids: list[str]
    segment_id: str # "home_online_sole_prop"
    area: str | None = None # registration | tax | employer
    requirement_ids: list[str] = []
    as_of: datetime
    limit: int = 5

class RetrievedChunk(BaseModel):
    chunk_id: str
    source_id: str
    text: str
    title: str
    section_path: str | None
    url: str
    source_version: str
    effective_from: datetime | None
    effective_to: datetime | None
    score: float | None

class RetrievalResult(BaseModel):
    status: Literal["supported", "insufficient_evidence"]
```

```
chunks: list[RetrievedChunk]
query_used: str
filters_applied: dict[str, Any]
limitations: list[str]
```

The retrieval service returns evidence only, never a legal conclusion. No one creates a second vector store or asks Gemini to answer from general knowledge.

7. Requirement registry

The registry is the checklist the whole system works from: about 15 to 18 reviewed requirements, written by the research owner in a shared sheet and loaded into the TiDB `requirements` table by a script.

Why a sheet: reading laws and writing requirements is research, not code. A sheet lets the business side write and review rows in parallel; `backend/scripts/load_registry.py` loads it into TiDB and is re-run whenever the sheet changes.

7.1 Fields

Field	Example	Notes
<code>id</code>	<code>TAX-01</code>	Format: <code>AREA-NN</code>
<code>area</code>	<code>tax</code>	<code>registration</code> , <code>tax</code> or <code>employer</code> ; decides which agent owns it
<code>title</code>	Register for BC PST	Plain English
<code>requirement_type</code>	<code>legal_obligation</code>	Or <code>recommendation</code> (never scored)
<code>timing</code>	<code>trigger</code>	<code>now</code> , <code>trigger</code> or <code>recommendation</code>
<code>applies_if</code>	<code>{"all":["sells in [goods, both]"]}</code>	Simple JSON rule over confirmed facts
<code>trigger_rule</code>	<code>pst_small_seller_test</code>	Which calculator decides it, if any
<code>required_fact_keys</code>	<code>sells</code> , <code>monthly_revenue</code> , <code>has_established_premises</code>	Missing ones become follow-up questions
<code>depends_on</code>	<code>REG-03</code>	Requirements that must come first
<code>priority</code>	<code>high</code>	<code>high</code> = 3, <code>medium</code> = 2, <code>low</code> = 1 in the

		score
source_chunk_ids	bc-pst-small-sellers-s2	Evidence that supports it
action_url	Official page or form	The only place links come from
preparation_items	Gather sales records for the past 12 months	Checklist shown to the owner
review_flags	Market stalls and premises	Known gray areas for the agent
last_verified_at	2026-10-03	When the research owner last checked it

7.2 Starter rows (all need research owner validation)

ID	Requirement	Timing	Applies when	Depends on
REG-01	Register business name with BC Registries	now	Trading under a name other than own legal name	none
REG-02	Get a City of Vancouver business licence	now	Operating in Vancouver, incl. from home (verify home-based and online-only)	none
REG-03	Get a CRA business number	trigger	Needed before GST or payroll accounts	none
TAX-01	Register for BC PST	trigger	Sells taxable goods and fails small seller test	none
TAX-02	Register for GST	trigger	Fails small supplier test	REG-03
TAX-03	Consider voluntary GST registration	recommendation	Below threshold with significant business purchases	REG-03
TAX-04	Charge and show the correct tax on invoices once registered	trigger	Registered for PST or GST	TAX-01 or TAX-02
EMP-01	Register with WorkSafeBC	trigger	First hire	none
EMP-02	Open a CRA payroll account	trigger	First hire	REG-03
EMP-	Pay at least \$18.25/hour	trigger	First hire	none

03				
EMP-04	Give a pay statement every payday	trigger	First hire	none
EMP-05	Keep payroll records for the required period	trigger	First hire (verify retention period)	none
EMP-06	Check employee vs contractor status	recommendation	Paying anyone for regular work	none

7.3 Fact dictionary (profile)

Fact	Type	Example	Used by
legal_name	text	Maya Chen	Registration
trading_name	text	Wick & Co	Registration
trading_name_differs_from_legal_name	bool	true	Registration
operates_in_vancouver	bool	true	Registration
home_based	bool	true	Registration, Tax
online_only	bool	false	Registration
sells	goods / services / both	both	Tax
monthly_revenue	list of month + amount	12 entries	Tax calculators
has_established_premises	bool	false	Tax (PST test)
sells_at_recurring_markets	bool	true	Tax (flag)
has_employees	bool	false	Employer
plans_to_hire	bool	true	Employer
planned_hire_date	month	2026-12	Employer

Every fact is stored as `{value, confirmed}`. Unknown or unconfirmed facts are never treated as false.

8. Front end, design brief and API contract

The designer builds from Figma (then into code with Claude Opus) against the API shapes below, so the screens match real data from day one.

8.1 Screens

1. **Intake:** one question per screen, plus an optional "describe your business" box.
2. **Confirm facts:** proposed facts as editable chips (confirm, edit, skip).
3. **Results dashboard:**
 - Large score ring with three area subscores (Registration, Tax, Employer).
 - Three columns or tabs: **Now, Next, Later**.
 - Action cards: priority colour, one-line reason, cost or time note, "Go here" button, "Ask about this", "Draft email".
 - Expandable "Why?" with the cited official passage and source link.
 - "Agent trace" panel showing what each agent checked.
 - Threshold progress bars for PST (\$10,000) and GST (\$30,000) with the projected crossing month.
4. **Chat:** side panel with cited answers and "confirm this change" prompts for proposed fact updates.
5. **Disclaimer:** visible "general information, not legal advice" note.

8.2 API routes (Developer 2 owns)

Route	In	Out
POST /intake/parse	Free-text description	Proposed facts with confidence
POST /profile	Confirmed facts + monthly revenue	Profile ID + version
GET /profile/{id}/questions		Follow-up questions for missing facts
POST /assess/{profile_id}		Assessment: overall score, area scores, now/next/later lists, findings
GET /requirements/{id}		Requirement details, citation, action link
GET /assessments/{id}/trace		Agent trace entries
POST /chat	Question + profile ID	Grounded answer, claims, sources, proposed fact update
POST /profile/{id}/confirm-update	Proposed update ID	New profile version, triggers re-assessment
POST /draft-email	Requirement ID + profile	Draft subject and body

8.3 Example assessment response (share with designer)

```
{
  "assessment_id": "a-102",
  "profile_version": 3,
  "score": 67,
  "area_scores": {"registration": 50, "tax": 100, "employer": null},
  "now": [
    {"requirement_id": "REG-01", "title": "Register your business name", "status":
"not_done",
      "priority": "high", "explanation": "You trade as Wick & Co, which is not your
legal name...",
      "action_url": "https://...", "sources": [{"title": "...", "url":
"https://..."}]}],
  "next": [
    {"requirement_id": "TAX-01", "title": "Register for BC PST", "status":
"not_yet_required",
      "progress": {"current": 8200, "threshold": 10000, "estimated_crossing": "2027-
03"}}],
  "later": [
    {"requirement_id": "EMP-01", "title": "Register with WorkSafeBC", "trigger":
"first_hire"}],
  "flags": [{"requirement_id": "TAX-01", "reason": "Regular market sales may affect
the premises condition"}],
  "disclaimer": "General information, not legal advice."
}
```

Example values are illustrative only. Store a full set of examples in `contracts/examples/`.

9. Task breakdown by owner

The critical path is registry content, then retrieval, then agents, then the dashboard; the research owner and both developers start in parallel in hour one.

9.1 Roles

Role	Owns
PM	Task assignment, rules check, demo script, pitch,

	.tech domain, UNSDG framing
Research owner (business)	Sources, requirement registry content, fact dictionary, threshold rules, gray areas, test questions
Developer 1: knowledge and assessment engine	TiDB, ingestion, chunking, embeddings, retrieval, registry loader, applicability rules, calculators, scoring, sequencing
Developer 2: agents and application	Intake, orchestrator, three agents and tools, validation, chat, actions, profile versioning, API routes, trace logging
Designer	Figma screens, front end build, deployment of front end

9.2 Phases

- 1. Phase 0, setup (first 2 hours):** TiDB cluster in a supported region, Gemini key and quota check, repo skeleton, shared contracts, first source end to end.
- 2. Phase 1, foundations (rest of day 1):** registry draft, ingestion, retrieval, calculators, intake, base agent class, Figma screens.
- 3. Phase 2, agents and engine (day 2 morning):** three agents, orchestrator, scoring, sequencing, API routes, front end wired to example JSON.
- 4. Phase 3, integration and demo (day 2 afternoon):** end-to-end Maya flow, chat and fact update loop, polish, deploy, rehearse.

9.3 Task list

ID	Task	Owner	Phase	Depends on
PM-01	Confirm StormHacks rules: track limits, pre-event design and research	PM	0	none
PM-02	Register .tech domain, set up DNS	PM	0	none
PM-03	Write demo script around Maya (section 3.2)	PM	2	RES-03
PM-04	Pitch deck: problem, comparison table, workflow, UNSDG framing, "same answers, same score" line	PM	3	none
PM-05	Run acceptance test checklist before demo	PM	3	all
RES-01	Finalise approved source list (section 6.2) with URLs	Research	0	none

RES-02	Confirm exact PST small seller and GST small supplier conditions from official pages	Research	1	RES-01
RES-03	Write and validate 15 to 18 registry rows in the shared sheet	Research	1	RES-01
RES-04	Resolve flagged items: Vancouver licence for home-based and online-only, licence fee, payroll record retention, market stalls	Research	1	RES-01
RES-05	Write 15 to 25 test questions, incl. at least 3 with no official answer	Research	1	RES-01
RES-06	Write Maya's full demo profile and 12 months of revenue	Research	1	none
D1-01	Create TiDB cluster in a full-text-supported region; verify vector and full-text search	Dev 1	0	none
D1-02	Create tables (section 4.2) and shared Pydantic contracts with Dev 2	Dev 1	0	D1-01
D1-03	Ingest first source end to end (fetch, chunk, embed, store, retrieve)	Dev 1	0	D1-02
D1-04	Ingest all approved sources with full metadata	Dev 1	1	RES-01, D1-03
D1-05	Retrieval service per contract (filters, vector or hybrid search, insufficient_evidence)	Dev 1	1	D1-03
D1-06	Registry loader script (sheet to TiDB)	Dev 1	1	D1-02
D1-07	Applicability engine: <code>applies_if</code> rules, statuses, missing facts	Dev 1	1	D1-06
D1-08	Calculators: PST small seller test, GST small supplier test, projected crossing	Dev 1	1	RES-02
D1-09	Scoring (section 2.5) and sequencing by <code>depends_on</code>	Dev 1	2	D1-07
D1-10	Retrieval evaluation against test questions	Dev 1	2	RES-05, D1-05
D2-01	FastAPI skeleton, settings, Gemini client, rate limit check	Dev 2	0	none
D2-02	Profile service with versioning and <code>{value, confirmed}</code> facts	Dev 2	1	D1-02
D2-03	Intake: free text to proposed facts (structured output)	Dev 2	1	D2-01
D2-04	Follow up question wording from missing facts	Dev 2	1	D1-07

D2-04	Follow-up question wording from missing facts	Dev 2	1	D1-07
D2-05	Base agent class: tools, two-phase loop, step limit, trace logging to <code>agent_runs</code>	Dev 2	1	D1-05
D2-06	Registration, Tax and Employer agents (prompts, scopes, tools, gray areas)	Dev 2	2	D2-05, RES-03
D2-07	Orchestrator: agent selection, parallel runs, merge	Dev 2	2	D2-06
D2-08	Citation validation	Dev 2	2	D2-06
D2-09	API routes (section 8.2) and example JSON in <code>contracts/examples/</code>	Dev 2	2	D1-09, D2-07
D2-10	Chat with agent routing, insufficient-evidence answers, fact update proposals	Dev 2	3	D2-07
D2-11	Inquiry email drafts	Dev 2	3	D2-09
D2-12	Cache Maya's assessment as demo fallback	Dev 2	3	D2-09
DES-01	Figma: intake, confirm facts, dashboard (now/next/later), card "Why?", trace, chat	Designer	1	none
DES-02	Build front end against example JSON	Designer	2	DES-01, D2-09
DES-03	Wire to live API, deploy to Vercel, connect domain	Designer	3	DES-02, PM-02

9.4 First joint session (30 minutes, before splitting up)

1. Pick one official source and one answerable question (for example: "Do I need to register Wick & Co?").
2. Extract three to five chunks and add metadata together.
3. Manually mark which chunk supports the answer.
4. Dev 1 walks through embed, store, filter, retrieve.
5. Dev 2 walks through agent tools, Gemini prompt, citation validation.
6. Test one unsupported question and agree on the insufficient-evidence wording.

Done when everyone can tell apart: poor retrieval, correct retrieval but unsupported answer, grounded answer, missing facts, wrong jurisdiction, outdated source.

10. Acceptance tests, non-goals, risks and open decisions

The MVP is done when the Maya flow runs end to end and every test below passes; anything not listed here is out of scope.

10.1 Acceptance tests

Business logic

- ☐ Owner trading under their own legal name gets no name registration requirement; "Wick & Co" does.
- ☐ \$8,200 rolling 12-month sales with no premises: PST shows as not yet required, with a projected crossing month.
- ☐ Crossing \$10,000 in a rolling 12 months moves PST registration to Now, even mid-year.
- ☐ A services-only owner does not get PST registration as required (outside the specified professional services).
- ☐ Missing revenue makes tax requirements undetermined, never "fine".
- ☐ Confirming a first hire moves all employer obligations from Later to Now and changes the score.
- ☐ Recommendations and upcoming items never change the score.
- ☐ Same profile always produces the same score.
- ☐ Next actions respect dependencies (business number before GST or payroll account).

Retrieval

- ☐ A Vancouver question returns Vancouver, BC or federal evidence, never another city's.
- ☐ Requirement filters restrict results correctly.
- ☐ Unsupported questions return `insufficient_evidence`.
- ☐ Every chunk has source metadata and a working URL.

Agents and chat

- ☐ Every compliance claim cites a chunk retrieved in that run.
- ☐ No invented fees, deadlines, forms or links.
- ☐ Gray areas appear as flags, not conclusions.
- ☐ Agent trace shows each tool call.
- ☐ A fact mentioned in chat becomes a proposed update, applied only after confirmation.

10.2 Non-goals for the MVP

- Incorporated businesses, partnerships, other cities or provinces.
- Food service, liquor, building permits, signage or other industry permits.
- Automated filings or sending emails automatically.
- Legal certification or advice.
- Continuous monitoring of law changes.

- Open web research during chat.
- Multi-agent debate (agents never argue; they each own their area and code merges results).
- A second vector store.

10.3 Risks and fallbacks

Risk	Fallback
TiDB full-text unavailable in our region	Use vector search only; still qualifies for the TiDB track
<code>pytidb</code> embedding wiring fails	Call Gemini embeddings directly and insert vectors via SQLAlchemy
Gemini rate limits during demo	Paid credits; cached Maya assessment
Registry not finished in time	Cut to 10 requirements (3 to 4 per area), keep the hiring moment
Legal facts wrong in front of judges	Research owner verifies every threshold and link; "general information, not legal advice" disclaimer visible
Dead action links	Link checker script run the night before

10.4 Open decisions for the team

- ☐ Exact source list and URLs (RES-01).
- ☐ Exact PST and GST test wording (RES-02).
- ☐ Vancouver licence treatment for home-based and online-only businesses (RES-04).
- ☐ Whether to use Google's Agent Development Kit or hand-rolled agent classes (default: hand-rolled, simpler to debug).
- ☐ Which StormHacks tracks we formally submit to (PM-01).

